

README for Structural Feasibility Confidence (SFC)

From Engineering Confidence to Structural Feasibility Metrics and the Structural Boundaries of LLMs

Sizhe Tan

with chatGPT (OpenAI)

2026-06-29

Repository: <https://github.com/sizhet/Structural-Feasibility-Confidence-SFC>

Structural Feasibility Confidence (SFC) proposes that engineering intelligence is fundamentally governed not by exact precedent matching, but by **structural feasibility evaluation**.

Human engineers routinely solve problems that have never appeared before. They succeed by judging whether a new structure is feasible, assembling partial evidence, estimating confidence, and continuously bridging gaps until a workable solution emerges.

This repository formalizes that process into a unified framework.

More importantly, it argues that many capabilities already exhibited by Large Language Models (LLMs) can be understood as implicit structural feasibility evaluation, providing a new engineering interpretation of modern LLMs while simultaneously clarifying their current structural limitations.

Repository Overview

Fig-016-SFC-DOI-Overall-Roadmap.png

The repository develops the theory through three progressively deeper stages.

Stage I

Structural Feasibility Confidence

↓

Stage II
Structural Feasibility Metrics

↓

Stage III
Structural Feasibility Scope

The progression moves from engineering practice, to mathematical formulation, and finally to architectural principles for future AI systems.

Stage I — Structural Feasibility Confidence

Fig-001-Confirmed-Zone-Extension-Zone-Speculative-Zone.png

This section explains **how engineers make feasible decisions without exact precedents.**

Topics include:

- Why exact precedent is insufficient
- Structural feasibility confidence
- Patchwork feasibility proof
- Function tunnel confidence thickness
- Task Calling Graphs as feasibility scaffolds
- Engineering commitment
- Gap Bridging
- LLM scoring as implicit confidence estimation

Rather than viewing engineering as retrieving identical examples, SFC models engineering as **incrementally constructing structural feasibility.**

Core documents include:

- SFC-001
- SFC-002
- SFC-003
- SFC-004
- SFC-005
- SFC-006
- SFC-007
- SFC-008
- SFC-009

Stage II — Structural Feasibility Metrics

Fig-013-From-Gap-Bridging-to-Feasible-Region.png

The second stage extends feasibility from a qualitative engineering judgment into a measurable structural framework.

Instead of asking

Is this possible?

the framework asks

How structurally feasible is it?

This section introduces:

- Structural Feasibility Metrics
- Feasibility Distance
- Autonomous Contracting
- Feasible Regions
- Structural Metrics beyond similarity

This represents a shift from traditional similarity-based evaluation toward **feasibility-oriented optimization**.

Core documents include:

- SFC-010
- SFC-010b
- SFC-011
- SFC-012
- SFC-013

Stage III — Structural Feasibility Scope

Fig-211-LLM-and-SlandASI-Component-and-Application-Coverage-Comparative-Analysis.png

The final stage investigates the **architectural boundaries of structural intelligence**.

Rather than asking whether a particular engineering solution is feasible, these papers ask:

What kinds of structural intelligence are possible under different internal representations?

This section provides an engineering interpretation of LLMs and explores

what additional structural components are required for future SI and ASI systems.

Topics include:

- LLMs as Implicit Structural Feasibility Engines
- Structural Feasibility Scope Principle (SFSP)
- Structural Collaborative Intelligence
- Structural Runtime Transfer Principle
- Structural Capability Closure Principle

These papers form the theoretical bridge between today's LLMs and future Structural Intelligence architectures.

Note:

The document numbers **SFC-210 through SFC-214** belong to this third stage. They are not a **separate series**; the numbering was intentionally shifted into the 2xx range solely to avoid conflicts with the existing SFC-010 through SFC-013 document numbers.

Core documents include:

- SFC-210
- SFC-210X
- SFC-212
- SFC-213
- SFC-214

Central Contributions

This repository makes several connected contributions.

1. Structural Feasibility as an Engineering Principle

Engineering is modeled as the continuous evaluation and construction of structural feasibility rather than exact precedent retrieval.

2. Structural Feasibility Metrics

Similarity is generalized into structural feasibility metrics capable of evaluating partially completed structures, evolving designs, and engineering trajectories.

3. LLMs as Implicit Structural Feasibility Engines

The repository proposes that many modern LLM behaviors—including scoring, ranking, preference optimization, and next-token prediction—can be understood as **implicit structural feasibility evaluation** rather than explicit symbolic reasoning.

This interpretation provides an engineering explanation for many practical strengths of LLMs.

4. Structural Feasibility Scope Principle (SFSP)

The repository introduces the **Structural Feasibility Scope Principle**, which distinguishes problems that can be solved through implicit structural feasibility evaluation from those requiring explicit structural representations such as runtime state, task graphs, differential trees, structural memory, or persistent internal representations. This principle offers an engineering framework for understanding both the strengths and limitations of current LLM architectures.

Fig-213-Structural-Runtime-Transfer-Principle.png

5. Toward Structural Intelligence

The repository argues that future Structural Intelligence (SI) and Artificial Super Intelligence (ASI) systems will require explicit structural runtime components extending beyond today's implicit feasibility mechanisms.

Relationship to Other Structural Intelligence Repositories

This repository forms part of the broader Structural Intelligence (SI) research program.

Within that framework:

Repository	Primary Focus
Structural Cognitive Emergence (SCE)	How structural cognition emerges
Structural Cognitive Runtime (SCR)	How structural cognition operates continuously
Structural Feasibility Confidence (SFC)	How structural feasibility is evaluated and constructed
Function Tunnel Intelligence (FTI)	How feasible evolution is preserved
Task-Action Dual Calling Graphs	Task-level structural organization
Universal Semantic Infrastructure (USI)	Shared semantic infrastructure
Universal Task Language (UTL)	Shared task representation

SFC serves as the missing engineering layer connecting structural cognition, runtime reasoning, optimization, and next-generation AI architectures.

Intended Audience

This repository is intended for researchers and practitioners interested in:

- Engineering reasoning
- Structural Intelligence
- Large Language Models
- AI Agents
- Runtime architectures
- Autonomous engineering
- AI system design
- Human-AI collaborative engineering

Reading Guide

Recommended reading order:

Part I

SFC-001 → SFC-009

Understanding engineering feasibility.

Part II

SFC-010 → SFC-013

Understanding feasibility metrics and optimization.

Part III

SFC-210 → SFC-214

Understanding the structural scope and future evolution of LLMs and Structural Intelligence.

Why This Repository Matters

The rapid success of Large Language Models has demonstrated that remarkably strong intelligence can emerge without explicitly programmed symbolic reasoning, handcrafted rules, or manually constructed world models.

However, one fundamental question remains largely unanswered:

What kind of intelligence are LLMs actually implementing?

This repository proposes a structural engineering interpretation rather than a purely statistical or cognitive one.

Instead of viewing LLMs primarily as language prediction systems, we argue that many of their most impressive engineering capabilities can be understood as **implicit structural feasibility evaluation**. During inference, LLMs continuously estimate whether partial structures are coherent, compatible, and extendable, allowing them to synthesize solutions that have

never existed as exact precedents.

This perspective provides a unified explanation for phenomena including ranking, scoring, next-token prediction, preference optimization, patchwork reasoning, engineering intuition, and many forms of practical problem solving.

The repository then asks a second, even more fundamental question:

Where does the structural feasibility capability of current LLMs begin to reach its architectural boundary?

The **Structural Feasibility Scope Principle (SFSP)** proposes that this boundary is determined not by model size alone, but by the availability of appropriate internal structural representations (IR).

Problems whose structural constraints can be evaluated implicitly are often well suited to today's LLM architectures. In contrast, problems requiring explicit runtime state, persistent structural memory, task graphs, differential structures, long-horizon structural evolution, or continuously maintained internal representations increasingly demand additional architectural components beyond implicit language modeling.

Rather than viewing this as a limitation of LLMs, SFSP provides a constructive engineering framework for understanding **which problems naturally match current architectures and which motivate the evolution toward future Structural Intelligence (SI) and Artificial Super Intelligence (ASI) systems.**

Taken together, **SFC-210** (LLMs as Implicit Structural Feasibility Engines) and **SFC-210X** (Structural Feasibility Scope Principle) establish a unified engineering interpretation of both the **strengths** and the **structural boundaries** of modern LLMs.

Importantly, the purpose of this repository is not to replace existing interpretations of Large Language Models. Rather, it proposes Structural Feasibility as an additional engineering coordinate system through which the capabilities, limitations, and future evolution of LLMs may be understood. Statistical learning, mechanistic interpretability, scaling laws, world models, and structural feasibility need not be viewed as competing explanations. Instead, they may illuminate different aspects of the same evolving engineering reality. The goal of SFC is therefore to complement existing perspectives by introducing a structural engineering language capable of connecting human engineering practice, LLM behavior, and future Structural Intelligence architectures.

Instead of framing the discussion as LLMs versus future architectures, this repository views them as successive stages along the same structural

engineering continuum:

**Implicit Structural Feasibility → Explicit Structural Intelligence →
General Structural Runtime Intelligence**

Importantly, the purpose of this repository is not to replace existing interpretations of Large Language Models. Rather, it proposes Structural Feasibility as an additional engineering coordinate system through which the capabilities, limitations, and future evolution of LLMs may be understood.

Statistical learning, mechanistic interpretability, scaling laws, world models, and structural feasibility need not be viewed as competing explanations. Instead, they may illuminate different aspects of the same evolving engineering reality. The goal of SFC is therefore to complement existing perspectives by introducing a structural engineering language capable of connecting human engineering practice, LLM behavior, and future Structural Intelligence architectures.

Under this perspective, today's LLMs are not regarded as an endpoint, but as the first large-scale engineering realization of structural feasibility computation, providing a practical foundation for the next generation of SI and ASI architectures.

This repository proposes that Structural Feasibility, rather than similarity, prediction, or symbolic manipulation alone, may serve as the missing engineering principle that connects modern LLMs with the next generation of Structural Intelligence architectures.

Whether Structural Feasibility ultimately proves to be a foundational principle remains an open scientific question.

Nevertheless, we believe that exploring AI through an additional structural engineering coordinate system may broaden both our theoretical understanding and our practical ability to design future intelligent systems. We therefore welcome empirical validation, critical discussion, alternative interpretations, and future improvements from the broader research community.

Author

Sizhe Tan

Independent Researcher

GPT-Obot

AI Research Assistant

2026

Citation

DOI: TBD

License

Apache-2.0



DBM-SI Series Navigation

See:

[./docs/DBM-SI-Series-of-github-Repositories/DBM-SI-Series-of-github-Repositories.md](#)

[./docs/DBM-SI-Series-of-github-Repositories/DBM-SI-Structural-Intelligence-Dictionary-\(v2\).md](#)